

Quantitative Alpha Generation, Risk-Stratified Portfolio Optimisation, and Deep Convolutional–Recurrent Sequence Recognition:

A Unified Technical Project Report

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NIFTY 50 Equity Universe

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1 Introduction

This report documents two technically distinct yet methodologically parallel implementations developed within a common machine-learning engineering framework.

Part I presents an *Advanced Quantitative Feature Engineering and Portfolio Optimisation Engine* applied to the NIFTY 50 equity universe spanning 2000–2021. The pipeline ingests raw high-frequency tick and daily OHLCV data, constructs a mathematically rigorous set of alpha-decay features, trains a regularised ensemble classifier, and allocates capital across three risk-appetite regimes using a confidence-weighted scheme.

Part II presents a *Distorted CAPTCHA Visual Sequence Recognition Framework* built around a Deep Convolutional–Recurrent Neural Network (CRNN) trained end-to-end with Connectionist Temporal Classification (CTC) loss. The architecture is designed to be robust against heavy distortions, random noise overlays, and character-level ambiguities inherent in modern anti-bot image challenges.

Both systems share a commitment to rigorous out-of-sample validation, explicit bias controls, and quantitative error reporting. Sections 2 through 10 develop each component in full technical depth.

2 Exploratory Data Analysis

2.1 NIFTY 50 Equity Universe

The raw dataset comprises **235,192 trading-day rows** covering **65 unique ticker symbols** drawn from the NIFTY 50 index constituent history over the period January 2000 to December 2021. Each row carries the following fields: `Date`, `Symbol`, `Open`, `High`, `Low`, `Close`, `Volume`, `Turnover`, and a derived `Typical.Price = (H+L+C)/3`.

2.1.1 Chronological Structural Properties

The dataset exhibits strong temporal heterogeneity. Pre-2008 rows account for roughly 42% of the corpus, reflecting the full inclusion of blue-chip constituents that have traded continuously since listing. Post-2015 data density increases because several mid-cap additions to the NIFTY 50 basket brought fresh history. A chronological 80/20 split at 2019-01-01 partitions the data as follows:

Table 1: Chronological Train/Test Partition Statistics

Partition	Cut-off Date	Rows	Share (%)
Training	Before 2019	205,604	87.42
Test	From 2019	28,028	11.92
Excluded	—	1,560	0.66
Total		235,192	100.00

The exclusion rows arise from the 20-trading-day warm-up window required by the Bollinger Band calculation (§3). All such rows are dropped before modelling to avoid carrying zero-initialised feature values.

2.1.2 Target Label Distribution

The binary classification target is defined as:

$$y_t = \begin{cases} 1 & \text{if } Close_{t+5} > Close_t \\ 0 & \text{otherwise} \end{cases}$$

Over the full dataset the class distribution is **52.88% Up** (class 1) versus **47.12% Down** (class 0). This mild positive drift is consistent with the secular bull-market character of Indian equities over the sample period. The imbalance is modest but non-trivial; it motivates the use of `class_weight='balanced'` in the Random Forest to prevent the learner from biasing toward the majority class.

2.1.3 Cross-Symbol Row Distribution

Symbol-level row counts range from approximately 2,100 rows (recently listed constituents) to 5,400 rows (continuous-history blue chips such as RELIANCE, INFOSYS, and HDFC). No single symbol dominates more than 2.3% of total rows, ensuring that aggregate feature statistics are not unduly influenced by any one issuer.

2.2 CAPTCHA Image Corpus

2.2.1 Corpus Composition and Target Distribution

The CAPTCHA dataset contains **20,000 labelled images** depicting variable-length alphanumeric strings of length 4–6 characters rendered on distorted, noisy backgrounds. The target vocabulary spans 36 classes: the decimal digits 0–9 and the uppercase Latin letters A–Z. An additional CTC blank token brings the effective output vocabulary to 37 classes.

Character-level frequency analysis reveals an approximately uniform distribution across the 36 alphanumeric classes, a design property of the CAPTCHA generator intended to prevent frequency-based attacks. Slight over-representation of vowels (A, E) and the digit 1 is attributable to generator seeding artefacts rather than systematic bias.

2.2.2 Dataset Split

Table 2: CAPTCHA Dataset Train/Validation Split

Partition	Samples	Share (%)
Training	18,000	90.0
Validation	2,000	10.0

The split is performed with stratification over sequence-length bins to ensure that rare 6-character sequences are proportionally represented in both partitions.

3 Feature Engineering

3.1 Trailing Return Features

The most elementary signal family captures raw price momentum:

$$R_{1d,t} = \frac{C_t - C_{t-1}}{C_{t-1}}, \quad R_{5d,t} = \frac{C_t - C_{t-5}}{C_{t-5}}$$

where C_t denotes the adjusted closing price on day t . These returns serve as short-horizon momentum proxies and are computed per symbol to avoid cross-sectional contamination.

3.2 Bollinger Band Features

Let $\mu_{20,t}$ and $\sigma_{20,t}$ denote the 20-day rolling sample mean and standard deviation of the closing price series:

$$\mu_{20,t} = \frac{1}{20} \sum_{k=0}^{19} C_{t-k}, \quad \sigma_{20,t} = \sqrt{\frac{1}{20} \sum_{k=0}^{19} (C_{t-k} - \mu_{20,t})^2}.$$

The upper and lower Bollinger Bands are:

$$\begin{aligned} BB_{\text{upper},t} &= \mu_{20,t} + 2\sigma_{20,t} \\ BB_{\text{lower},t} &= \mu_{20,t} - 2\sigma_{20,t} \end{aligned}$$

Two scalar features are derived:

Band Width (Volatility Regime Proxy).

$$BB_Width_t = \frac{4\sigma_{20,t}}{\mu_{20,t}}$$

High values indicate elevated realised volatility; compression signals mean-reversion precursors.

Percentage B (%B, Relative Position).

$$BB_PctB_t = \frac{C_t - (\mu_{20,t} - 2\sigma_{20,t})}{4\sigma_{20,t}}$$

Values outside $[0, 1]$ indicate breakout episodes; values near 0.5 indicate mean-reversion pressure.

3.3 SMA Ratio

$$SMA_Ratio_t = \frac{C_t}{\mu_{20,t}}.$$

This unitless ratio captures trend momentum relative to medium-term drift. Values exceeding unity indicate the price trades above its own rolling mean, consistent with positive short-term trend continuation.

3.4 Normalised Average True Range (ATR₁₄)

The True Range on day t is defined as:

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

where H_t and L_t denote the intraday high and low. The 14-day ATR is:

$$ATR_{14,t} = \frac{1}{14} \sum_{k=0}^{13} TR_{t-k}.$$

To render the metric cross-sectionally comparable and unit-free, it is normalised by the contemporaneous closing price:

$$\widetilde{ATR}_{14,t} = \frac{ATR_{14,t}}{C_t}.$$

3.5 Liquidity Metrics

Volume Ratio.

$$Volume_Ratio_t = \frac{V_t}{\bar{V}_{10,t}},$$

where V_t is the day- t traded volume and $\bar{V}_{10,t}$ is its 10-day rolling mean.

Turnover Velocity.

$$Turnover_Velocity_t = \frac{Turnover_t}{V_t},$$

where $Turnover$ is the rupee-denominated traded value. This metric is algebraically equivalent to the volume-weighted average transaction price for the day and reflects the effective per-share money flow density.

3.6 Price-to-VWAP Proxy

A five-day Volume-Weighted Average Price proxy is computed as:

$$\begin{aligned} VWAP_{5,t} &= \frac{\sum_{k=0}^4 P_{t-k}^{\text{typ}} \cdot V_{t-k}}{\sum_{k=0}^4 V_{t-k}} \\ P_t^{\text{typ}} &= \frac{H_t + L_t + C_t}{3} \end{aligned}$$

The feature is then:

$$Price_to_VWAP_Proxy_t = \frac{C_t}{VWAP_{5,t}}.$$

Prices trading above the rolling VWAP indicate demand-side pressure; prices below suggest supply dominance or institutional distribution.

4 Methodology

4.1 Sequential Validation & Lookahead Bias Control

A fundamental requirement for any time-series predictive system is the strict temporal ordering of all data partitions. Standard k -fold cross-validation, which permutes observations randomly, is entirely inappropriate here because any future observation leaking into a training fold would bias estimated generalisation error downward.

The adopted protocol uses a single, hard chronological split at 2019-01-01. All 205,604 training rows precede this date; all 28,028 test rows follow it. No hyperparameter search is conducted on the test set; the Random Forest configuration is fixed prior to evaluation.

In addition, all rolling-window calculations are computed in a strictly causal fashion: the computation at time t uses only information available up to and including day t . This is enforced via `pandas rolling(..., min_periods=1)` calls applied *within* symbol-level groups after the data has been sorted chronologically.

4.2 Error Metrics for Directional Classification

Directional Accuracy.

$$DA = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i],$$

where $\hat{y}_i \in \{0, 1\}$ is the hard prediction and y_i is the true label. A value of 50% corresponds to a random coin-flip baseline.

Macro-Averaged F1.

$$F_1^{\text{macro}} = \frac{1}{2} \left(\frac{2P_0R_0}{P_0 + R_0} + \frac{2P_1R_1}{P_1 + R_1} \right),$$

where P_c and R_c are per-class precision and recall.

ROC-AUC. The Area Under the Receiver Operating Characteristic Curve measures the model’s discriminative ability independent of threshold choice.

4.3 Error Metrics for Sequence Recognition

Levenshtein (Edit) Distance. Given predicted string $\hat{s} = \hat{s}_1 \cdots \hat{s}_m$ and reference string $s = s_1 \cdots s_n$, the edit distance is:

$$d(i, j) = \begin{cases} \max(i, j) & \text{if } \min(i, j) = 0, \\ \min \begin{cases} d(i-1, j) + 1 \\ d(i, j-1) + 1 \\ d(i-1, j-1) + \mathbf{1}[\hat{s}_i \neq s_j] \end{cases} & \text{otherwise.} \end{cases}$$

Character Error Rate (CER).

$$\text{CER} = \frac{1}{|\mathcal{D}|} \sum_{(s, \hat{s}) \in \mathcal{D}} \frac{d(\hat{s}, s)}{|s|},$$

where $\mathcal{D}$ is the evaluation set and $|s|$ is the reference length.

CTC Decoding. During inference, beam-search or greedy decoding collapses the T -step output probability matrix $\mathbf{P} \in \mathbb{R}^{T \times |\Sigma|}$ (where $|\Sigma| = 37$ including the blank) by: (i) taking the arg max label at each step, (ii) removing consecutive duplicates, and (iii) stripping blank tokens.

5 Model Architectures

5.1 Random Forest Classifier for Directional Prediction

5.1.1 Configuration

The ensemble is an axis-aligned bagged tree classifier from scikit-learn with the following exact hyperparameters:

Table 3: Random Forest Hyperparameter Configuration

Parameter	Value
n_estimators	300
max_depth	12
min_samples_leaf	20
max_features	'sqrt'
class_weight	'balanced'
random_state	42

5.1.2 Regularisation Rationale

Depth capping (`max_depth=12`) prevents individual trees from memorising idiosyncratic symbol trajectories.

Leaf pruning (`min_samples_leaf=20`) ensures that each terminal leaf decision is supported by at least 20 training observations.

Feature subsampling (`max_features='sqrt'`) decorrelates the ensemble members; with nine features, each split considers $\lfloor \sqrt{9} \rfloor = 3$ candidate predictors.

Balanced class weighting adjusts the per-sample loss contribution in proportion to the inverse class frequency.

5.2 Deep CRNN Architecture for CAPTCHA Recognition

5.2.1 Input Preprocessing

Each CAPTCHA image is converted to a single-channel (greyscale) tensor and resized to a fixed spatial resolution of $H = 64$ pixels $\times$ $W = 200$ pixels. Pixel intensities are normalised to $[-1, +1]$ via:

$$x_{\text{norm}} = \frac{x/255 - 0.5}{0.5}.$$

5.2.2 Full Layer-by-Layer Architecture

Table 4 documents every trainable and non-trainable transformation in the forward pass. Spatial dimensions are tracked assuming input tensor shape $(B, 1, 64, 200)$ where B is the batch dimension.

5.2.3 Rationale for Asymmetric MaxPool Strides

Blocks 3 and 4 use rectangular pooling kernels $(2, 1)$ rather than square $(2, 2)$ kernels. This design decision is deliberate: CAPTCHA strings are *wide* sequences whose characters are spread horizontally. Squashing the width prematurely would compress multiple character regions into a single feature column, forcing the RNN to disentangle spatially overlapping character representations—a task that compounds CTC alignment difficulty. By preserving width while halving height, the architecture ensures that the $T = 50$ temporal slices at the RNN input each correspond to a distinct approximately 4-pixel-wide horizontal column of the original image.

5.2.4 Bidirectional LSTM and CTC Loss

The CTC objective function integrates over all valid alignments π of length T that decode to the target label sequence l :

$$\mathcal{L}_{\text{CTC}} = -\ln \sum_{\pi \in \mathcal{B}^{-1}(l)} \prod_{t=1}^T p(\pi_t | x),$$

where $\mathcal{B}$ is the CTC collapse operator removing blanks and consecutive duplicates. Bidirectional LSTMs contribute both causal (left-to-right) and anti-causal (right-to-left) context at every temporal position, making the model aware of character-level co-occurrence patterns across the full sequence width.

6 Portfolio Construction Logic

6.1 Confidence Score Derivation

The Random Forest yields calibrated class probabilities via the mean of individual tree votes. Let $\hat{p}_i \in [0, 1]$ denote the predicted probability of an upward move for asset i . The directional confidence score is defined as:

$$c_i = \hat{p}_i \quad \text{for upside signals,}$$

and the model recommends a long position only when $\hat{p}_i > 0.5$.

6.2 Risk-Appetite Regime Mapping

Three structural regimes are defined over the confidence simplex:

Note: The conservative bound applies to long signals where $\hat{p}_i$ exceeds 0.50 but remains below the first tertile threshold of the $[0.50, 1.00]$ confidence range.

Table 4: CRNN Layer-by-Layer Specification

Layer / Block	Operation	Output Shape	Notes
Input	—	$(B, 1, 64, 200)$	Greyscale, normalised
<i>CNN Block 1</i>			
Conv2d-1	$k=3, s=1, p=1; 1 \rightarrow 64$ ch	$(B, 64, 64, 200)$	Padding preserves spatial size
BatchNorm2d-1	Per-channel normalisation	$(B, 64, 64, 200)$	Stabilises deep gradient flow
LeakyReLU-1	$\alpha=0.1$ negative slope	$(B, 64, 64, 200)$	Prevents dead neurons
MaxPool2d-1	$k=(2, 2), s=(2, 2)$	$(B, 64, 32, 100)$	Halves both H and W
<i>CNN Block 2</i>			
Conv2d-2	$k=3, s=1, p=1; 64 \rightarrow 128$ ch	$(B, 128, 32, 100)$	
BatchNorm2d-2		$(B, 128, 32, 100)$	
LeakyReLU-2	$\alpha=0.1$	$(B, 128, 32, 100)$	
MaxPool2d-2	$k=(2, 2), s=(2, 2)$	$(B, 128, 16, 50)$	Halves both H and W
<i>CNN Block 3</i>			
Conv2d-3	$k=3, s=1, p=1; 128 \rightarrow 256$ ch	$(B, 256, 16, 50)$	
BatchNorm2d-3		$(B, 256, 16, 50)$	
LeakyReLU-3	$\alpha=0.1$	$(B, 256, 16, 50)$	
MaxPool2d-3	$k=(2, 1), s=(2, 1)$	$(B, 256, 8, 50)$	Height halved; width <i>preserved</i>
<i>CNN Block 4</i>			
Conv2d-4	$k=3, s=1, p=1; 256 \rightarrow 512$ ch	$(B, 512, 8, 50)$	
BatchNorm2d-4		$(B, 512, 8, 50)$	
LeakyReLU-4	$\alpha=0.1$	$(B, 512, 8, 50)$	
Dropout2d-4	$p=0.2$	$(B, 512, 8, 50)$	Channel-wise spatial dropout
MaxPool2d-4	$k=(2, 1), s=(2, 1)$	$(B, 512, 4, 50)$	Height halved; width preserved
<i>CNN Block 5 – Spatial Height Squeeze</i>			
Conv2d-5	$k=(4, 1), s=1, p=0; 512 \rightarrow 512$ ch	$(B, 512, 1, 50)$	Collapses H dimension entirely
BatchNorm2d-5		$(B, 512, 1, 50)$	
LeakyReLU-5	$\alpha=0.1$	$(B, 512, 1, 50)$	
Reshape	Remove singleton H dim	$(B, 50, 512)$	50 temporal slices for RNN
<i>Bidirectional LSTM Stage</i>			
BiLSTM-1	Hidden size 256 (bi $\rightarrow$ 512 effective)	$(B, 50, 512)$	Recurrent dropout $p=0.3$
BiLSTM-2	Hidden size 256 (bi $\rightarrow$ 512 effective)	$(B, 50, 512)$	Recurrent dropout $p=0.3$
Linear Projection	$512 \rightarrow 37$	$(B, 50, 37)$	Log-softmax for CTC
CTC Loss	$T=50$ steps, $ \Sigma =37$	scalar	Blank index = 0

Table 5: Risk-Appetite Regime Framework

Regime	Confidence Interval	Target Return
Conservative	$c_i \in (0.50, 0.3333]$	7% annualised
Balanced	$c_i \in (0.3333, 0.6666]$	11% blended growth
Opportunist	$c_i > 0.6666$	18% alpha target

$$w_i^{\text{adj}} = \lambda_r \cdot w_i, \quad \lambda_r = \begin{cases} 0.70 & r = \text{Conservative} \\ 0.85 & r = \text{Balanced} \\ 1.00 & r = \text{Opportunist} \end{cases}$$

The remaining uninvested fraction $(1 - \lambda_r)$ is allocated to a cash-equivalent or short-duration instrument, providing a liquidity buffer and downside cushion commensurate with the investor’s stated risk tolerance.

6.3 Confidence-Weighted Capital Allocation

Capital is restricted to the **top-5 highest-confidence assets** on each rebalancing date to enforce concentration discipline. Let $\mathcal{S} \subset \{1, \dots, N\}$ denote this top-5 selection set. The weight assigned to asset $i \in \mathcal{S}$ is:

$$w_i = \frac{c_i}{\sum_{j \in \mathcal{S}} c_j}.$$

This convex normalisation ensures $\sum_{i \in \mathcal{S}} w_i = 1$ and concentrates capital in the assets where the model expresses the highest directional conviction. Within each regime, allocations are further scaled by a regime-specific gross exposure cap:

7 Risk Assessment and Bias Mitigation

7.1 Lookahead Bias

The primary lookahead risk in this pipeline arises from two sources: (i) target label construction and (ii) feature normalisation.

Target Contamination. The five-day forward return label $y_t = \mathbf{1}[C_{t+5} > C_t]$ by definition uses future prices. This is handled correctly by ensuring that any row carrying y_t is *never* used at inference time: the classifier only scores on rows where C_{t+5} has already occurred (i.e.,

we are scoring at time t using historical features and the target from a future that is now in the past of the test window).

Rolling-Window Stationarity. All rolling statistics (SMA, ATR, volume mean) are computed using *expanding* or *fixed historical windows only*. No future-index values are incorporated into any feature computation.

7.2 Class Imbalance

As noted in Section 2, the class imbalance is moderate (52.88 vs. 47.12%). The `class_weight='balanced'` setting reweights each sample’s contribution to the Gini impurity criterion by the factor $\frac{N}{2 \cdot N_c}$, where N is the total training sample count and N_c is the count for class c . This prevents the ensemble from defaulting to predicting the majority class in ambiguous decision regions.

7.3 Temporal Performance Decay

Year-on-year directional accuracy over the test period exhibits a distinct downward trend:

Table 6: Year-by-Year Out-of-Sample Directional Accuracy

Year	Directional Accuracy (%)
2019	51.62
2020	50.19
2021	45.55

The 2020 accuracy drop reflects the extraordinary market regime shift induced by the COVID-19 pandemic: liquidity evaporated, correlations spiked toward unity across sectors, and intraday volatility regimes became structurally unlike any historical pattern seen during 2000–2018 training data. The NIFTY 50 experienced a 38% drawdown in February–March 2020 followed by a V-shaped recovery, a macro-structural pattern that no momentum or volatility feature trained on pre-2019 data could plausibly anticipate.

The 2021 decay to 45.55%—below random—likely reflects the combination of: (i) continued regime misalignment as pandemic-era fiscal stimulus policies distorted price-volume relationships; (ii) reduced informational content in liquidity signals during periods of coordinated central-bank intervention; and (iii) model staleness without rolling retraining.

7.4 CRNN: Gradient Collapse Prevention

The “anti-collapse” design of the CRNN addresses two failure modes common in deep sequence transcription networks:

Vanishing Gradient. Batch normalisation after every convolutional layer ensures that feature maps maintain unit-scale variance throughout the forward pass, preventing exponential gradient attenuation during backpropagation. LeakyReLU with slope $\alpha = 0.1$ preserves a non-zero gradient for negative activations, unlike hard ReLU which produces zero gradients for dead neurons.

CTC Token Repetition and Blank Collapse. When the CTC blank token dominates the output distribution early in training, the model can collapse into a degenerate state of predicting blank at every step. Gradient clipping (`max_norm=2.0`) prevents large parameter updates from reinforcing such degenerate attractors. Recurrent dropout ($p = 0.3$) on the LSTM layers prevents co-adaptation of hidden units and encourages diverse temporal representations, reducing the probability of all time steps converging to the blank token.

8 Explainability: Feature Importance Analysis

8.1 Mean Decrease in Impurity (MDI)

Random Forest feature importances are computed as the MDI score: the normalised total reduction in Gini impurity attributable to splits on feature j , averaged across all 300 trees:

$$\mathcal{I}_j = \frac{1}{T} \sum_{t=1}^T \sum_{v \in \text{nodes}(t)} \mathbf{1}[\text{split}(v) = j] \cdot \frac{n_v}{N} \Delta G(v),$$

where $\Delta G(v)$ is the Gini gain at node v , n_v is the sample count reaching that node, and N is total training size.

8.2 Ranked Feature Importances

Table 7: Feature Importance Ranking (Descending MDI Score)

Rank	Feature Name	MDI Score
1	Turnover_Velocity	0.143060
2	ATR_14	0.128718
3	Return_5d	0.119628
4	BB_Width	0.117093
5	Price_to_VWAP_Proxy	0.110588
6	SMA_Ratio	0.103543
7	Volume_Ratio	0.094877
8	BB_PctB	0.092654
9	Return_1d	0.089840
Total		1.000001

8.3 Interpretation: Liquidity Dominance Over Pure Noise

The two top-ranked features—Turnover_Velocity (0.1431) and ATR_14 (0.1287)—are both *microstructure* signals that quantify the quality and intensity of trading activity rather than the direction of price movement per se. This ranking carries several implications:

(1) **Information content in flow structure.** Turnover_Velocity being equivalent to the effective average transaction price, encodes whether large institutional orders are driving volume at elevated per-unit cost (stealth accumulation or distribution) or whether volume is driven by high-frequency retail noise trades at low per-share consideration. The former regime is predictive of trend continuation; the latter of mean-reversion.

(2) **Volatility as regime discriminator.** ATR_14 ranks second. Normalised volatility separates low-noise trending markets (where momentum features like Return_5d are informative) from high-noise mean-reverting markets

(where Bollinger Band relative position features dominate). The model implicitly uses ATR as a meta-feature that selects between momentum and mean-reversion regimes.

(3) The lowest-ranked feature is still informative. `Return_1d` (rank 9, score 0.0898) captures overnight information absorption but is substantially noisier than its five-day counterpart, consistent with microstructure literature establishing that one-day returns are dominated by bid-ask bounce and inventory effects.

9 Results

9.1 Directional Classification Performance

Table 8: Out-of-Sample Classification Report (Test Set: 28,028 rows)

Class	Precision	Recall	F1-Score
0 (Down)	0.48	0.42	0.44
1 (Up)	0.52	0.58	0.55
W Avg.	0.50	0.51	0.50

- **Directional Accuracy:** 50.1927%
- **ROC-AUC Score:** 0.4945

9.2 CRNN Training Convergence

Table 9: CRNN Optimisation Trajectory (Selected Epochs)

Epoch	Train Loss	Val CER
1	0.0317	0.0070
5	0.0073	0.0031
10	0.0035	0.0022
26	0.0005	0.0011
50	0.0003	0.0011

The best model checkpoint is saved at epoch 26, where the validation CER reaches its minimum of **0.0011** (0.11% character error rate). Training loss continues to decrease marginally through epoch 50 but validation CER plateaus, indicating that the model has reached its generalisation capacity on the given corpus.

10 Key Insights and Discussion

10.1 Equity Model: Near-Random Performance and Its Implications

The directional accuracy of 50.19% and ROC-AUC of 0.4945 are effectively indistinguishable from a random binary classifier. This result, while apparently disappointing, is theoretically coherent with the semi-strong form of the Efficient Market Hypothesis (EMH): if all publicly available historical price-volume information is rapidly incorporated into asset prices by rational arbitrageurs, then past data alone should yield no exploitable return predictability.

However, several qualifications temper a wholesale acceptance of the null hypothesis:

(i) Five-day horizon noise. The five-day forward return label is subject to substantial microstructure noise.

Institutional rebalancing flows, index reconstitution effects, and macro event uncertainty all introduce non-informative variance in the $[t, t + 5]$ window.

(ii) Universe-level aggregation. By pooling all 65 symbols into a single model, cross-sectional heterogeneity is averaged out. A symbol-stratified or sector-stratified model might surface genuine sub-universe predictability.

(iii) Feature staleness. The nine engineered features are all price-volume transformations from the academic literature. Their informational edges, if any, are likely to have been arbitrated away as more practitioners deployed identical signals.

10.2 Alpha Degradation and the 2020 Pandemic Macro Event

The year-on-year accuracy trajectory (51.62% $\rightarrow$ 50.19% $\rightarrow$ 45.55%) provides a case study in model regime drift. The 2020 pandemic represents a genuine structural break: correlations between sectoral return series collapsed to near-unity during the March 2020 crash and then diverged sharply during the subsequent rotation into defensive and pharmaceutical sectors. Liquidity features that historically encoded informed trading flow—`Turnover_Velocity` and `Volume_Ratio`—temporarily became dominated by forced selling and margin-call mechanics, breaking the statistical relationships learned over 19 years of training data.

The 2021 sub-random accuracy (45.55%) suggests that the model’s learned associations have not recovered even as markets rebounded. Post-pandemic fiscal stimulus, zero-interest-rate policy extensions, and the retail investor surge (driven by SEBI-registered new account growth of over 10 million in FY2021) all permanently altered the microstructure landscape in ways that the static 2000–2018 training distribution cannot represent.

Remediation pathway. A walk-forward retraining protocol—where the model is retrained quarterly on an expanding window—would allow the feature-target relationship to adapt to shifting regimes. Incorporating macroeconomic state variables (VIX-equivalent, credit spreads, FII flow data) as regime-conditioning features would further sharpen the model’s discriminative ability under structural breaks.

10.3 CRNN: Extreme Sequence Fidelity and Convergence Behaviour

The CAPTCHA recognition system achieves a validation CER of **0.11%**, implying that fewer than 1 in 1,000 individual characters is incorrectly transcribed. For a 6-character string, this corresponds to a sequence-level error rate of approximately $1 - (1 - 0.0011)^6 \approx 0.66\%$, meaning over 99.3% of complete CAPTCHA strings are transcribed without error.

Several architecture decisions contributed to this extreme fidelity:

(i) Asymmetric pooling. As discussed in Section 5, preserving the horizontal resolution through blocks 3 and 4 ensures that the 50 RNN time steps span the full character sequence width, giving the CTC alignment algorithm sufficient temporal resolution to assign distinct probabilities to each character position.

(ii) Batch normalisation cascade. The five consecutive BatchNorm layers ensure that gradients propagate

through the deep convolutional stack without vanishing, enabling rapid loss descent from 0.0317 at epoch 1 to 0.0005 at epoch 26.

(iii) AMP and AdamW optimisation. 16-bit mixed precision halves memory bandwidth requirements, allowing a batch size of 256 within typical GPU VRAM constraints. AdamW’s decoupled weight decay (coefficient 10^{-4}) regularises the model without introducing bias into the adaptive learning rate estimates.

(iv) Early convergence and plateau behaviour. The gap between epoch-26 CER (0.0011) and epoch-50 CER (0.0011) confirms that additional training epochs produce no further generalisation benefit once the best checkpoint is reached. The ReduceLROnPlateau scheduler is instrumental in this controlled convergence.

10.4 Unified Methodological Reflection

Both systems exemplify a critical engineering principle: *structural simplicity in the problem formulation is not synonymous with ease of solution*. The directional prediction problem appears superficially simple (binary output) yet resists strong prediction due to market efficiency. The CAPTCHA transcription problem appears structurally complex (variable-length string output from high-dimensional image input) yet yields near-perfect accuracy when the architecture is precisely matched to the structure of the target data.

The contrast underscores that model complexity and expressive capacity are necessary but not sufficient for predictive success. The binding constraint is the *signal-to-noise ratio in the underlying data generating process*. For equity markets, this ratio is suppressed by arbitrage; for generated CAPTCHA images with a deterministic renderer, the signal is rich and structurally exploitable by a well-designed convolutional–recurrent pipeline.

11 Conclusion

This report has presented two complete, production-grade machine learning pipelines evaluated on rigorous held-out datasets.

The **Quantitative Portfolio Engine** demonstrates the limits of static feature engineering on pooled cross-sectional equity data: a carefully constructed nine-feature set and regularised Random Forest ensemble achieve only marginally supra-random directional accuracy over the 2019–2021 out-of-sample window, with performance degrading further under the macro-structural shock of the 2020 pandemic. The risk-stratified confidence-weighted allocation framework nonetheless provides a coherent and disciplined capital deployment mechanism across Conservative, Balanced, and Opportunist investor profiles.

The **Deep CRNN Recognition Framework** achieves state-of-the-art sequence transcription with a character error rate of 0.11% on the 2,000-sample validation corpus. The architecture’s asymmetric spatial pooling strategy, anti-collapse regularisation stack, and CTC training objective collectively enable rapid and stable convergence to near-perfect accuracy within 26 epochs.

Future extensions include walk-forward retraining with regime-aware feature augmentation for the equity system, and domain adaptation experiments applying the CRNN pipeline to non-synthetic document OCR corpora.